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FUEL INJECTOR DRIVE SYSTEM

Abstract

A fuel injector drive system may include a fuel injector having a first electronically-controlled valve and a second electronically-controlled valve. The fuel injector drive system may include a fuel injector driver configured to supply current to the first electronically-controlled valve and the second electronically-controlled valve simultaneously to actuate the first electronically-controlled valve and the second electronically-controlled valve. The fuel injector drive system may also include an injection valve configured to inject fuel when the first electronically-controlled valve and the second electronically-controlled valve are actuated.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates generally to methods and systems for internal combustion engine components and, more particularly, to systems and methods for a fuel injector with multiple electronically-controlled valves.

BACKGROUND

[0002] Fuel injectors for internal combustion engines are designed to inject a controllable amount of fuel. While some fuel injectors include a single electronically-controlled valve, such as a solenoid valve, other fuel injectors include multiple electronically-controlled valves. As an example, some injectors include a first valve that facilitates pressurization of fuel within the fuel injector and a second valve that facilitates injection of this pressurized fuel. These valves may be individually controlled to operate the separate valves of a particular fuel injector at different times. For example, a valve that enables fuel pressurization might be activated before or after a valve that controls injection is activated.

[0003] While injectors with individually-controlled valves offer advantages, these valves also increase the cost and complexity of the fuel injection system. For example, each electronically-controlled valve is connected to a dedicated drive circuit that supplies energy at appropriate times. Thus, some engine systems require two drive circuits for each fuel injector. To accommodate the numerous drive circuits, some engines include multiple separate control modules, these modules being in communication with each other to coordinate fuel injection events during the operation of the engine. While these systems are effective and can accurately operate an internal combustion engine, the use of multiple control modules increases the number of possible failure points and further increases cost. For example, some engines require double the number of injection drivers compared to the number of engine cylinders (e.g., a 16 cylinder engine may require 32 injection drivers).

[0004] An exemplary fuel injector control circuit is described in U.S. Pat. No. 5,701,870 (“the '870 patent”) to Gottshall et al. The fuel injector control circuit described in the '870 patent can be used with a plurality of different fuel injectors and can be programmed to produce a plurality of different injector current waveforms. The control circuit includes a microprocessor with memory connected to a multiplexer and an application-specific integrated circuit. The control circuit can also be used to increase the current rise time of a specific injector current waveform. However, the '870 patent does not describe connecting the injector actuators of a single fuel injector in series or in parallel, so that a single injector driver can control both solenoids and allow the injector to inject fuel in a controlled manner while reducing power.

[0005] The systems and methods of the present disclosure may solve one or more of the problems set forth above and/or other problems in the art. The scope of the current disclosure, however, is defined by the attached claims, and not by the ability to solve any specific problem.

SUMMARY

[0006] In one aspect, a fuel injector drive system may include a fuel injector having a first electronically-controlled valve and a second electronically-controlled valve. The fuel injector drive system may include a fuel injector driver configured to supply current to the first electronically-controlled valve and the second electronically-controlled valve simultaneously to actuate the first electronically-controlled valve and the second electronically-controlled valve. The fuel injector drive system may also include an injection valve configured to inject fuel when the first electronically-controlled valve and the second electronically-controlled valve are actuated.

[0007] In another aspect, a fuel injection system may include a fuel injector having a first valve and a second valve, a fuel injector drive circuit electrically connected to the first valve of the fuel injector and to the second valve of the fuel injector to supply simultaneous current for both the first valve and for the second valve, and an electronic control module configured to generate a command for energizing the fuel injector drive circuit and generate a command for actuating the

first valve and the second valve.

[0008] In still another aspect, a fuel injection method may include supplying current to a first electronically-controlled valve with an injector driver, supplying current to a second electronically-controlled valve with the injector driver simultaneously as the current is supplied to the first electronically-controlled valve, and injecting fuel via an injection valve of a fuel injector that may include the first electronically-controlled valve and the second electronically-controlled valve when the first electronically-controlled valve and the second electronically-controlled valve actuate in response to the current.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is a diagram of a cylinder arrangement, according to aspects of the disclosure.

[0010] FIG. 2 is a schematic cross-sectional view of a fuel injection system, according to aspects of the disclosure.

[0011] FIG. 3 is a diagram illustrating a first example of a drive circuit for paired fuel injector valves useful in the fuel injection system of FIG. 2, according to aspects of the disclosure.

[0012] FIG. 4 is a diagram illustrating a second example of a drive circuit for paired fuel injector valves useful in the fuel injection system of FIG. 2, according to aspects of the disclosure.

[0013] FIG. 5 is a chart showing exemplary current waveform for the system of FIG. 1, according to aspects of the disclosure.

[0014] FIG. 6 is a flowchart depicting an exemplary fuel injection method, according to aspects of the disclosure.

DETAILED DESCRIPTION

[0015] Both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the features, as claimed. As used herein, the terms “comprises,” “comprising,” “having,” “including,” or other variations thereof, are intended to cover a non-exclusive inclusion such that a method or apparatus that comprises a list of elements does not include only those elements, but may include other elements not expressly listed or inherent to such a method or apparatus. In this disclosure, relative terms, such as, for example, “about,” “substantially,” “generally,” and “approximately” are used to indicate a possible variation of $\pm 10\%$ in the stated value or characteristic. In this disclosure, the phrase “based on” encompasses both “based at least on” and “based entirely on.”

[0016] FIG. 1 illustrates an exemplary engine system **10**, according to aspects of the present disclosure. Engine system **10** may be installed in an internal combustion engine. Engine system **10** may include a plurality of cylinders (**62, 64, 66, 68, 70, and 71**). Each of the plurality of cylinders **62, 64, 66, 68, 70, and 71** may include a respective fuel injector (**72, 74, 76, 78, 80, and 82**) that is controlled via a respective injector driver **54**, as described below. Each of the fuel injectors may be electrically connected to an electronic control module (ECM) **52**. ECM **52** may be electrically connected to a high-voltage power system (HVPS) **58** and to a battery **60**. While FIG. 1 is illustrated with six cylinders, one of ordinary skill in the art will appreciate that the aspects of the present disclosure may be applicable to a variety of engines, including engines with four, six, eight, ten, twelve, or twenty cylinders.

[0017] FIG. 2 illustrates an exemplary fuel injector drive system **12** of engine system **10**, according to aspects of the present disclosure. Fuel injector drive system **12** may include fuel injector **72** and ECM **52** that includes a plurality of injector drivers (e.g., injector drivers **54**) that are electrically connected to valves of fuel injector **72**. Fuel injector **72** may be installed in an internal combustion engine having a plurality of engine cylinders as described with respect to FIG. 1. In some aspects, fuel injector drive system **12** may be one of a plurality of fuel injector drive systems designed to be

identical to each other (e.g., fuel injector drive system **12** and the plurality of fuel injector drive systems may be structurally identical) and may be fuel injector drive systems of the same type, such as fuel injector drive systems having the same model number. Each of the plurality of fuel injector drive systems may correspond with a cylinder of an internal combustion engine having a plurality of cylinders. It should be understood that while injector **72** is shown and described as having two solenoids corresponding to two valves, injector **72** may have additional solenoids corresponding to additional valves (e.g., three solenoids corresponding to three valves).

[0018] Fuel injector **72** and fuel injectors **74**, **76**, **78**, **80**, and **82** may be designed to be identical to each other (e.g., one or more of fuel injectors **72**, **74**, **76**, **78**, **80**, and **82** may be structurally identical) and may be fuel injectors of the same type, such as fuel injectors having the same model number. Therefore, the description of fuel injector **72** also applies to fuel injectors **74**, **76**, **78**, **80**, and **82**, except when indicated otherwise.

[0019] Fuel injector **72** may be a mechanically-actuated electronically-controlled unit injector including a body that houses a camshaft-driven piston (not shown), a fuel passage **20** to receive fuel pressurized when this piston is pressed by the camshaft, an electronically-controlled spill valve **22**, an electronically-controlled control valve **26**, and an injection valve **30**. Spill valve **22** may be a normally-open valve including a valve member **24** that is movable between an open position and a closed position. A spring member may act to bias spill valve member **24** to the open position. When the valve member **24** is in the open position, spill valve **22** may allow fuel to drain and return to the fuel supply system. When in the closed position, spill valve **22** may enable pressurization of fuel via the piston of fuel injector **72**. Spill valve **22** may include a spill valve solenoid **42** for actuating spill valve member **24** due to movement of a spill valve armature **46** to which valve member **24** is connected. Spill valve solenoid **42** may be energized in response to commands from ECM **52**, the energized state acting to move spill valve **22** to the closed position.

[0020] Control valve **26** may be connected between pressurized fuel supply passage **20** and a control chamber **38**. Control valve member **28** may have a non-injection position and an injection position. When in the non-injection position, control valve **26** may enable fluid communication between control chamber **38** and fuel that is pressurized with a piston, filling chamber **38** with fuel. When control valve member **28** is in the injection position, control chamber **38** may be depressurized by allowing fuel in chamber **38** to drain from fuel injector **72** to the low pressure fuel supply and drain system. Control valve **26** may be brought to the injection position due to electromagnetic force created by supplying current to control valve solenoid **44**.

[0021] Injection valve **30** may be a one-way valve formed with a spring, an injection valve member **32** biased by the spring, and control chamber **38**. An upper hydraulic surface of injection valve member **32** may face control chamber **38**. When high-pressure fluid is present in control chamber **38**, injection valve member **32** may be secured in a closed position, even when pressurized fuel is present in injection chamber **34**, due to the ability of fluid within control chamber **38** to block movement of injection valve member **32**. When injection is desired, fluid may be permitted to drain from control chamber **38**, as described below, allowing pressurized fuel to lift injection valve member **32** by acting on a lower hydraulic surface of injection valve member **32**.

[0022] ECM **52** may be an electronic control module that controls one or more aspects of fuel injection drive system **12**, including the behavior of an internal combustion engine. ECM **52** may be implemented as a single control unit that controls multiple aspects of system **12** and, in particular, control of every fuel injector present in system **12** (e.g., fuel injector **72** in the internal combustion engine of system **12**). ECM **52** may be enabled, via programming, to generate commands that control fuel injection events, including commands for a fuel injector drive circuit (also referred to herein as a “fuel injector driver” or “injector driver”) that includes solenoids of a fuel injector. ECM **52** may also be configured, via programming, to monitor performance of valves of the fuel injector and, based on fuel injector currents, determine when valves connected to a fuel injector driver operate abnormally (e.g., due to an open circuit, a short circuit, valve sticking, etc.).

ECM 52 may be configured to generate a notification based on an identified abnormality, this notification identifying the fuel injector in which the abnormality occurred. If desired, ECM 52 may also be configured to monitor valve return times based on current induced by motion of spill valve member 24 and, in some embodiments, control valve member 28. Based on the valve return time of spill valve member 24, for example, ECM 52 may be configured to adjust timing of valve 22 in one or more injection events.

[0023] ECM 52 may embody a single microprocessor or multiple microprocessors that receive inputs and generate outputs. ECM 52 may include a memory, a secondary storage device, a processor such as a central processing unit, or any other means for accomplishing a task consistent with the present disclosure. The memory or secondary storage device associated with ECM 52 may store data and software to allow ECM 52 to perform its functions, including the functions described with respect to fuel injection method 600, described below. Numerous commercially available microprocessors can be configured to perform the functions of ECM 52. Various other known circuits may be associated with ECM 52, including signal-conditioning circuitry, communication circuitry, drive circuitry, power-source command circuitry, and other appropriate circuitry. ECM 52 may include circuitry for controlling a single injector driver to simultaneously supply power to a pair of solenoid valves of a respective fuel injector, as described below. ECM 52 may include circuitry for HVPS 58 configured to boost a voltage of electrical energy applied to solenoids 42 and 44. While HVPS 58 is shown outside of ECM 52, HVPS 58 may instead be implemented within ECM 52. Further, while injector driver 54 is shown as a component of ECM 52, in at least some configurations, injector driver 54 is partially or entirely external to ECM 52. In some embodiments, ECM 52 may be configured to control the application of energy to solenoids 42 and 44 via battery 60 and HVPS 58. ECM 52 may issue commands to selectively energize solenoids 42 and 44 with electrical power from battery 60 and/or HVPS 58, and de-energize solenoids 42 and 44 to control a rate of decay of electrical energy (e.g., current through coils of solenoids 42 and 44).

[0024] ECM 52 may also be configured to detect valve return times based on current induced in the injector drive circuit (e.g., driver 54). For example, the return of spill valve member 24 at the end of actuation may induce current, a peak of this current corresponding to the return time. This return time of valve member 24 may occur after control valve member 28 has returned and after the termination of current generated with command 102. Thus, induced current within the drive circuit after the termination of command 102 may correspond to valve member 24, allowing ECM 52 to compare a detected return timing to a desired return timing, as described below. Based on the detected return timing, ECM 52 may modify one or more future commands 102 to provide improved control over spill valve 22.

[0025] FIG. 3 is a diagram illustrating an exemplary drive circuit 54 for driving fuel injector 72. The drive circuit 54 may include a voltage source 100 (e.g., HVPS 58 and/or battery 60) configured to receive a voltage command 102, a first solenoid (e.g., control valve solenoid 44) having a first resistance 104 and a first inductance 108, and a second solenoid (e.g., spill valve solenoid 42) having a second resistance 106 and a second inductance 110. The drive circuit 54 may also include a ground potential 112. Electrical components for two valves of a single fuel injector, such as solenoids 42 and 44 for spill valve 22 and control valve 26, respectively, may be electrically connected in series with each other as shown in FIG. 3.

[0026] Electrically connecting the injector actuators (e.g., solenoids 42 and 44) in series may allow a single injector driver (e.g., injector driver 54) to control both solenoids 42 and 44 simultaneously and allow the injector driver 54 to inject fuel in a controlled manner while staying under a power supply limit. This simultaneous control may result in current being supplied in both solenoids 42 and 44 for the same amount of time for every fuel injection event. However, characteristics of the two valves, such as one valve being more quickly actuated than the other when supplied with the same quantity of current, may cause the valves to be actuated for different periods of time when exposed to the same amount of energy simultaneously.

[0027] Voltage source **100** may be responsive to voltage commands **102** issued by ECM **52**. These commands may selectively open and close the drive circuit such that, when the drive circuit is closed, voltage source **100** is connected and current flows through both solenoid **42** and solenoid **44**. When both solenoids **42** and **44** are acting in an intended manner, the flow of current may energize both solenoids simultaneously.

[0028] Current through solenoids **42** and **44** may be impacted by resistance **104**, resistance **106**, inductance **108**, and inductance **110**. The flow of current, and in particular, the rise time of current within the drive circuit during the application of an approximately constant voltage, may be monitored to facilitate diagnostics of the valves associated with solenoids **42** and **44**.

[0029] FIG. **4** is a diagram illustrating another exemplary drive circuit **54** for driving valves of fuel injector **72**. As described above, drive circuit **54** may include a voltage source **100** configured to receive a voltage command **102**, a first solenoid **44**, a second solenoid **42**, and a ground potential **112**. The electrical elements illustrated in FIG. **4** may feature any of the features or characteristics of the electrical elements described in FIG. **3**.

[0030] FIG. **4** depicts solenoids **44** and **42** connected electrically in parallel. This parallel connection may also allow a single driver **54** to provide electrical energy to both solenoids **44** and **42** simultaneously, in the manner described above. One of ordinary skill in the art will appreciate that the parallel arrangement of solenoids **44** and **42** as shown in FIG. **4** may require additional current as compared to the series arrangement illustrated in FIG. **3**. Therefore, voltage source **100** may be scaled upward as necessary for actuating electronically-controlled spill valve **22** and electronically-controlled control valve **26** of a single injector **72**.

INDUSTRIAL APPLICABILITY

[0031] System **12** may be useful in any internal combustion engine, such as liquid fuel (e.g., diesel, gasoline, etc.) engines, gaseous fuel engines, or dual-fuel engines (engines configured to combust two types of fuels). For example, a dual-fuel engine may combust a gaseous fuel (e.g., natural gas) and a liquid fuel (e.g., diesel fuel), a first liquid fuel (e.g., diesel fuel) and a second liquid fuel (e.g., methanol), or a first gaseous fuel and a second gaseous fuel. System **12** may be utilized for generating power in a stationary machine (e.g., a generator or other electricity-generating device), in a mobile machine (e.g., an earthmoving device, a hauling truck, a drilling machine, etc.), or in other applications in which it is beneficial to install a plurality of fuel injectors in an internal combustion engine and connect two or more injector solenoids such that each fuel injector is driven by a single injector driver.

[0032] During an injection event, the pressure of fuel within pressurized fuel passage **20** (FIG. **1**) may increase when spill valve **22** is closed and a camshaft drives the piston within fuel injector **72** downward. Control valve **26** may control whether fluid (e.g., fuel) within control chamber **38** is pressurized.

[0033] When fuel injection is desired in a particular cylinder (e.g., cylinder **62**) of system **12**, ECM **52** may generate a drive circuit command **102** to a drive circuit **54** for the injector (e.g., injector **72**) installed in that cylinder. For example, ECM **52** may generate voltage command **102** (FIGS. **3** and **4**) to supply electrical energy to a first valve of injector via injector driver **54** (FIG. **1**). This energy may actuate the first valve of fuel injector **72**, spill valve **22**, and move spill valve member **24** of fuel injector **72** to the closed position.

[0034] The same command **102** may cause valve member **28** of control valve **26** of fuel injector **72** to move to the injection position. This actuation may allow fluid within control chamber **38** to drain via a low-pressure fluid passage (e.g., a fluid drain). Pressurized fuel within injection chamber **34** of fuel injector **72** may then lift injection valve member **32** and permit injection of fuel. It should be noted that command **102**, while referred to as a single command, may include a plurality of instructions (e.g., a plurality of commands) and may cause a plurality of actions to be performed by components electrically connected to a drive circuit (e.g., drive circuit **54**).

[0035] FIG. **5** is a chart showing an exemplary waveform **500** that may represent an amount of

current through a drive circuit (e.g., injector driver **54**) electrically connected to spill valve solenoid **42** and control valve solenoid **44** (e.g., in series as shown in FIG. 3). Waveform **500** may represent an amount of current (y-axis) supplied through both spill valve solenoid **42** and control valve solenoid **44** over time (x-axis) during an injection event. As used herein, an “injection event” refers to an injection of fuel, directly or indirectly, to a combustion chamber, and includes a main injection, a pilot injection, and/or a post injection. In some aspects, an injection event includes at least a main injection.

[0036] Waveform **500** may begin with a start of current **502**. Ramp-up current **504** may be applied for a period of time beginning at start of current **502** and ending at the beginning of a pull-in portion **506**.

[0037] Waveform **500** may include, in addition to ramp-up current **504**, a pull-in portion **506**, a first hold-in portion **508**, and a second hold-in portion **510**. Pull-in portion **506** may include a maximum current present in waveform **500**. Hold-in portion **508** of waveform **500** follows pull-in portion **506** and may include one or more current tiers that are smaller than the hold-in current. In the example illustrated in FIG. 5, hold-in portion **508** includes an intermediate current level between pull-in portion **506** and hold-in portion **510**.

[0038] Hold-in portion **510** of waveform **500** follows hold-in portion **508** and may include one or more current tiers that are smaller in magnitude than hold-in portion **508**. In the example illustrated in FIG. 5, pull-in portion **506**, hold-in portion **508** and hold-in portion **510** includes “chopped” current levels that include repeated minima and maxima, but this is not required. A chopped current may be a current that is interrupted or modulated at specific current amplitudes, times, or other intervals. While waveform **500** is illustrated at different amplitudes (heights) in FIG. 5, these heights are exemplary, with a current of zero amps being represented by the beginning of waveform **500** on the left-hand side of FIG. 5 showing the beginning the waveform **500**.

[0039] As understood, the amplitude of the current supplied to spill valve solenoid **42** and control valve solenoid **44** may be correlated with the amount of force generated to pull and hold spill valve member **24** and control valve member **28**, respectively, in their actuated positions. A higher pull-in current associated with pull-in-portion **506** may generate a greater amount of force, for example, to cause spill valve member **24** and control valve member **28** to each overcome static friction and other forces to move to the closed position and injection position, respectively. Lower levels of current (e.g., hold-in current **508**, **510**) may generate lower forces. In many instances, these forces may be sufficient to maintain spill valve **22** and control valve **26** in their respective actuated positions.

[0040] Following end of current **511**, energy in solenoid **44** may generate a reduced amount of magnetic force that no longer holds spill valve member **24** in the open position. When spill valve member **24** returns from the actuated position (closed or injection position) to the resting position (open or non-injection position), this movement may generate, via induction, a freewheeling current having a peak **512**. This freewheeling current peak **512** corresponds to the timing at which spill valve member **24** returns to the resting position (e.g., by contacting a valve seat associated with the open, non-injection position). The time between end of current **511** and current peak **514**, forms a valve return time, the amount of time that elapses between the end of the current and the actual return of the valve member, spill valve member **24**, to a corresponding seat.

[0041] FIG. 6 shows a flowchart illustrating an exemplary fuel injection method **600** for injecting fuel, according to aspects of the disclosure. It should be understood that in some aspects, at least part of the method **600** may be commands issued by ECM **52** of system **10**. In some aspects, a command may contain one or more instructions, and each of the one or more instructions may cause one or more actions to be performed by one or more elements electrically connected to drive circuit **54**, as described above. For example, voltage command **102** generated by ECM **52** may contain instructions to cause the steps **602** and **604** to be performed by one or more elements electrically connected to drive circuit **54**.

[0042] A first step **602** of method **600** may include supplying current to a first electronically-controlled valve with an injector driver (e.g., drive circuit **54**). This may correspond with start of current **502** as illustrated in FIG. 5. For example, ECM **52** may generate voltage command **102** that causes current to be supplied to electronically-controlled spill valve **22** (via solenoid **42**). As noted above, electronically-controlled spill valve **22** may be electrically connected to electronically-controlled control valve **26** in series or in parallel.

[0043] A second step **604** of method **600** may include supplying current to a second electronically-controlled valve with the same injector driver (e.g., drive circuit **54**), simultaneously as the current is supplied to the first electronically-controlled valve. For example, ECM voltage command **102** may cause current to be supplied to electronically-controlled control valve **26** (via solenoid **44**). Energizing (e.g., supplying current to) the solenoid **42** and the solenoid **44** may actuate the first electronically-controlled valve and the second electronically-controlled valve at different times, although in some embodiments the actuation of the two valves begins at the same time. Steps **602** and **604** may therefore be performed entirely contemporaneously for a single injector **72** associated with a single drive circuit **54**.

[0044] A third step **606** of method **600** may include injecting fuel via an injection valve when the first electronically controlled valve and the second electronically controlled valve actuate due to the simultaneous supply of current for valves **22** and **26**. For example, injection valve **30** may inject fuel after the spill valve **22** and the control valve **26** are actuated in steps **602** and **604**.

[0045] A fourth step **608** may be performed continuously or intermittently during method **600**. Step **608** is optional and is not included in all embodiments. During step **608**, ECM **52** may monitor a fuel injector (e.g., fuel injector **72**) installed in an internal combustion engine to identify issues, such as faults, in particular spill and control valves (e.g., spill valve **22** and control valve **26**) of the injector. For example, ECM **52** may be configured to monitor current rise times when command **102** is generated. A current rise time may be associated with a particular fault or issue. Similarly, ECM **52** may also be configured to monitor induced current peaks **512** after command **102** is no longer generated. For example, induced current peaks **512** that occur later than expected after end of current **511** may be indicative of a particular fault or issue, such as a sticking valve. ECM **52** may include circuitry for communicating via wireless and/or wired connections to a display device within system **12** or at a remote location, so as to enable ECM **52** to cause display of notifications. For example, ECM **52** may determine that a current rise time was abnormally fast. ECM **52** may generate and transmit a notification based on the abnormally fast rise time. The transmitted notification of the abnormally fast rise time may be shown to a user via a display, for example.

[0046] In some aspects, the disclosed system and method may enable a single injector driver to control two solenoids to control and power engines with a relatively large number of fuel injectors, such as an engine with 16 fuel injectors, 20 fuel injectors, or more. The system may realize cost reduction by the reduction in the required number of injector drivers (e.g., by including half the number of injector drivers as compared to fuel injectors), and may improve reliability by avoiding the need to provide communication connections, which may become loose, damaged, or even severed, during operation.

[0047] It will be apparent to those skilled in the art that various modifications and variations can be made to the disclosed system and method without departing from the scope of the disclosure. Other embodiments of the system and method will be apparent to those skilled in the art from consideration of the specification and system and method disclosed herein. It is intended that the specification and examples be considered as exemplary only, with a true scope of the disclosure being indicated by the following claims and their equivalents.

Claims

- 1.** A fuel injector drive system, comprising: a fuel injector having: a first electronically-controlled valve; and a second electronically-controlled valve; a fuel injector driver configured to supply current to the first electronically-controlled valve and the second electronically-controlled valve simultaneously to actuate the first electronically-controlled valve and the second electronically-controlled valve; and an injection valve configured to inject fuel when the first electronically-controlled valve and the second electronically-controlled valve are actuated.
- 2.** The fuel injector drive system of claim 1, further comprising: a first solenoid that, when energized, actuates the first electronically-controlled valve; and a second solenoid that, when energized, actuates the second electronically-controlled valve.
- 3.** The fuel injector drive system of claim 2, wherein the first solenoid and the second solenoid are electrically connected in series to each other.
- 4.** The fuel injector drive system of claim 2, wherein the first solenoid and the second solenoid are electrically connected in parallel to each other.
- 5.** The fuel injector drive system of claim 2, wherein the fuel injector driver is configured to energize the first solenoid and the second solenoid simultaneously.
- 6.** The fuel injector drive system of claim 5, wherein energizing the first solenoid and the second solenoid actuates the first electronically-controlled valve and the second electronically-controlled valve at different times.
- 7.** The fuel injector drive system of claim 1, wherein the first electronically-controlled valve is a spill valve configured to control pressure of fuel within the fuel injector.
- 8.** The fuel injector drive system of claim 1, wherein the second electronically-controlled valve is an injection control valve.
- 9.** The fuel injector drive system of claim 1, wherein the fuel injector drive system includes a plurality of fuel injectors and a plurality of fuel injector drivers, a quantity of fuel injectors being equal to a number of fuel injector drivers.
- 10.** A fuel injection system, comprising: a fuel injector having a first valve and a second valve; a fuel injector drive circuit electrically connected to the first valve of the fuel injector and to the second valve of the fuel injector to supply simultaneous current for both the first valve and for the second valve; and an electronic control module configured to: generate a command for energizing the fuel injector drive circuit; and generate a command for actuating the first valve and the second valve.
- 11.** The fuel injection system of claim 10, further comprising: a first solenoid that, when energized, actuates the first valve; and a second solenoid that, when energized, actuates the second valve.
- 12.** The fuel injection system of claim 11, wherein the first solenoid and the second solenoid are electrically connected in series.
- 13.** The fuel injection system of claim 11, wherein the first solenoid and the second solenoid are electrically connected in parallel.
- 14.** The fuel injection system of claim 10, wherein the fuel injection system is a first fuel injection system of a plurality of fuel injection systems, each of the plurality of fuel injection systems corresponding to a cylinder of a plurality of cylinders of an internal combustion engine.
- 15.** The fuel injection system of claim 14, wherein a quantity of fuel injectors and a quantity of fuel injector drive circuits of the plurality of fuel injection systems are both equal to a quantity of cylinders of the plurality of cylinders.
- 16.** A fuel injection method, comprising: supplying current to a first electronically-controlled valve with an injector driver; supplying current to a second electronically-controlled valve with the injector driver simultaneously as the current is supplied to the first electronically-controlled valve; and injecting fuel via an injection valve of a fuel injector that includes the first electronically-controlled valve and the second electronically-controlled valve when the first electronically-controlled valve and the second electronically-controlled valve actuate in response to the current.

17. The fuel injection method of claim 16, further including: detecting one or more conditions of the first electronically-controlled valve or the second electronically-controlled valve based on current supplied via the injector driver.

18. The fuel injection method of claim 16, wherein the injector driver that simultaneously supplies current to the first electronically-controlled valve and to the second electronically-controlled valve is a single injector driver.

19. The fuel injection method of claim 18, wherein the first electronically-controlled valve and to the second electronically-controlled valve are electrically connected in series to each other.

20. The fuel injection method of claim 16, wherein the first electronically-controlled valve of the fuel injector is a spill valve and the second electronically-controlled valve of the fuel injector is a control valve.
